CS 210: Worksheet 18 *Dr. Mudassir Shabbir, October 27, 2026*

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Functions, with a review of posets. Adapted from Rosen, *Discrete Mathematics and Its Applications*, 7th ed., § 2.3 and § 9.6.

1. Well-defined functions *Easy*

For each relation/rule below, explain why it does not define a function on $\mathbb{R}$.

(a) $f(x) = 1/x$.

(b) $f(x) = \sqrt{x}$.

(c) $f(x) = \pm\sqrt{x^2 + 1}$.

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2. Domain and range *Easy*

Find the domain and range of each function on finite bit strings, including the empty string.

(a) The number of 1s minus the number of 0s.

(b) Twice the number of 0s.

(c) The number of bits left over after grouping the string into complete bytes of 8 bits each.

3. Injections and Surjections *Easy*

For each function $\mathbb{Z} \to \mathbb{Z}$, decide whether it is injective (one-to-one) and whether it is surjective (onto). Prove each positive claim; for each negative claim, give a collision or a missing output.

(a) $f(n) = n - 1$.

(b) $f(n) = n^2 + 1$.

(c) $f(n) = n^3$.

(d) $f(n) = \lceil n/2 \rceil$.

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4. Composition *Moderate*

Let $f, g : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^2 + 1$ and $g(x) = x + 2$.

(a) Compute the composition $(f \circ g)(x)$.[1]

(b) Compute $(g \circ f)(x)$.

[1] The *composition* $f \circ g$ applies g first, then f: $(f \circ g)(x) = f(g(x))$.

5. Bijection, Invertibility and the domain *Easy*

Consider $f : \mathbb{R} \to [0, \infty)$ given by $f(x) = |x|$.

(a) Show that f is onto but is not invertible.[2]

(b) Restrict the domain to $[0, \infty)$. Prove that the resulting function is invertible and give its inverse, including its domain and codomain.

6. Composition preserves properties *Challenge*

Let $g : A \to B$ and $f : B \to C$. Use the definitions to prove each statement about their composition $f \circ g$.[3]

(a) If f and g are both one-to-one, then $f \circ g$ is one-to-one.

(b) If f and g are both onto, then $f \circ g$ is onto.

[2] A function $f : A \to B$ is *invertible* if every $b \in B$ has exactly one preimage $a \in A$ with $f(a) = b$; equivalently, f is bijective. Its *inverse* $f^{-1} : B \to A$ sends b to that unique a, undoing f. The notation f^{-1} does not mean $1/f$.

[3] For $g : A \to B$ and $f : B \to C$, the *composition* $f \circ g : A \to C$ is defined by $(f \circ g)(a) = f(g(a))$: apply g first, then f.

7. A divisibility Hasse diagram *Easy*

Let $S = \{2, 3, 5, 10, 11, 15, 25\}$, ordered by divisibility. Draw the Hasse diagram, identify all minimal and maximal elements, and determine whether this poset has a least element or a greatest element.

8. Upper and lower bounds *Moderate*

Work in the divisibility poset $(S, |)$, where $S = \{3, 5, 9, 15, 24, 45\}$. All bounds must belong to S. Compare bounds using divisibility.

(a) Find all upper bounds of $\{3, 5\}$ and its least upper bound, if one exists.[4]

(b) Find all lower bounds of $\{15, 45\}$ and its greatest lower bound, if one exists.[5]

[4] In a poset $(S, \preceq)$, an *upper bound* of $A \subseteq S$ is a $u \in S$ with $a \preceq u$ for every $a \in A$. A *least upper bound* is an upper bound u with $u \preceq v$ for every upper bound v. In the divisibility order, upper bounds are common multiples in S; the least upper bound divides every upper bound.

[5] A *lower bound* of $A \subseteq S$ is an $\ell \in S$ with $\ell \preceq a$ for every $a \in A$. A *greatest lower bound* is a lower bound ℓ with $v \preceq \ell$ for every lower bound v. In the divisibility order, lower bounds are common divisors in S; the greatest lower bound is divisible by every lower bound.